SEAN CARTER

GEOSPATIAL DATA SCIENTIST

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Geospatial data scientist specializing in satellite remote sensing and machine learning for earth systems modeling. Expertise in processing geospatial datasets and developing deep learning models for land use change, snow parameter estimation, and otherwise, across continental scales.

EDUCATION

Master of Science, Systems Ecology 2022
University of Montana (Missoula, MT)
Thesis: *Facilitating Aquatic Invasive Species Management Using Satellite Remote Sensing and Machine Learning Frameworks*

Bachelor of Arts, Organismal Biology and Ecology 2016
Minor: Classical Piano Performance
Colorado College (Colorado Springs, CO)

APPOINTMENTS AND EXPERIENCE

Professional and Research Appointments

Earth Intelligence, Leidos
Geospatial Data Scientist (2025 – Present)
- Built SpatioTemporal Asset Catalog indexing tool on terrabyte-scale data across cloud-based platforms
- Designed snow modeling system on integrated passive microwave, radar, and optical satellite datasets

Working Lands for Wildlife, University of Montana
Research Scientist (2023 – 2025)
- Developed an annual deep learning UNET model achieving 90% accuracy in identifying cultivated land use change across a cumulative 260 million square kilometers
- Directed multi-terabyte data engineering effort to visualize land-use change across the western United States since 1950

RegenIntel
Research Consultant (2025 – Present)
- Edited, managed, and provided professional input on carbon verification methodologies for international action-based farming operations

Weatherman Foundation
Data Science Consultant (2025 – Present)
- Analyzed and presented clinical trial data on Stellate Ganglion Block treatments for PTSD

Spatial Analysis Lab, University of Montana
Remote Sensing Consultant (2023)
- Developed temporally explicit land-use change alert system for invasive annual grass *(Ventenata dubia)*

OneConcern
Supply Chain Data Scientist Consultant (2022 – 2023)
- Developed supply chain resilience framework using internal SQL pipelines and digital twin methodology

Bay Area Environmental Research Institute
Data Scientist (2022)
- Designed, prototyped, and implemented a data fusion modeling framework with GEO-LEO imagery for fire detection in near real-time
- Integrated cloud computing software (AWS) with operational model output to process high fidelity geostationary imagery data

Numerical Terradynamic Simulation Group, University of Montana
Research Assistant (2020 – 2022)
- Formulated a probabilistic risk framework for biological invasions by fusing multi-temporal satellite remote sensing data, in-situ sensors, and field-based measurements

Field and Internship Experience

Bird Ecology Lab, University of Montana
Riverine Avian Field Biologist (2019)

Bird Conservancy of the Rockies
Avian Field Biologist (2018)

Southwestern Research Station, American Museum of Natural History
Avian Field Technician (2017 – 2018)

Central Plains Experimental Range, Agricultural Research Service
Avian Field Technician (2016)

Dauphin Island Sea Lab, University of South Alabama
Research Assistant (2015)

Appleton-Whittell Research Ranch, National Audubon Society
Grassland Field Technician (2014)

Music Experience

Sin Rumbo
Band Leader (2020 – Present)

Rocky Horror Picture Show, The Wilma
Musical Accompanist (2025)

Rocky Horror Picture Show, The Wilma
Performance Pianist (2022 – 2024)

Miscellaneous Performances, Western Montana's LGBTQ+ Community Center
Performance Pianist (2023 – 2025)

PUBLICATIONS

2025 **S. C. Carter.,** S.L. Morford, J. Tack, T. Lark, N. Uludere Aragon, B.W. Allred, D. Twidwell, D.E. Naugle. (In Press.) "Annual cultivated extent and agricultural land use expansion across the Central Grasslands of North America, 1996–2021." *Ecological Indicators*. https://doi.org/10.1101/2025.03.18.643874

Howard, L., C.B. van Rees, **S. C. Carter**, W. Daniel, B.K. Hand, M. Neilson, A. Sepulveda, R. Valavi, J.S. Kimball, G. Luikart. (In Review.) "Improving rapid assessment of invasion risk with Big Data on aquatic invasive species occurrence and remotely sensed environmental variables." *Management of Biological Invasions*.

2022 van Rees, C.B., B.K. Hand, **S. C. Carter**, C. Bargeron, T.J. Cline, W. Daniel, J.A. Ferrante, K. Gaddis, M.E. Hunter, C.S. Jarnevich, M.A. McGeoch, J.T. Morisette, M.E. Neilson, H.E. Roy, M.A. Rozance, A. Sepulveda, R.D. Wallace, D. Whited, T. Wilcox, J.S. Kimball, G. Luikart. (2022.) "A framework to integrate innovations in invasion science for proactive management." *Biological Reviews* 97(4): 1712–1735. https://doi.org/10.1111/brv.12859

2021 **S. C. Carter**, C.B. van Rees, B.K. Hand, C.C. Muhlfeld, G. Luikart, J.S. Kimball. (2021.) "Testing a generalizable machine learning workflow for aquatic invasive species on Rainbow Trout (*Oncorhynchus mykiss*) in Northwest Montana." *Frontiers in Big Data* 4. DOI: 10.3389/fdata.2021.734990. https://doi.org/10.3389/fdata.2021.734990

Estes, M.G. Jr., R.H. Carmichael, X. Chen, **S. C. Carter.** (2021.) "Environmental factors and occurrence of horseshoe crabs in the northcentral Gulf of Mexico." *PLOS ONE* 16(1): 1–22. https://doi.org/10.1371/journal.pone.0243478

SOFTWARE AND APPLICATIONS

Cropland Extent Mapping Application. Interactive web application.
Available at: https://wlfw-um.projects.earthengine.app/view/cropland-extent-map

Landscape Explorer. Historical land-use visualization from 1950s to present.
Available at: *www.landscapeexplorer.org*

Annual Cropland Extent Analysis Pipeline. Code repository (for demonstration purposes only). Github.
Available at*:* https://github.com/COYE-Coder/Annual-Cropland-Extent/tree/main

CONFERENCE PRESENTATIONS

2024 **S. C. Carter**., Morford, S.L., Lark, T.J., Uludere Aragon, N.Z., Tack, J., Naugle, D.E. (2024, December 13). CropScope: A high-resolution (30-m), annual

cropland extent product for North America's Grasslands (1996–2021) [Poster presentation, Session GC51K-0102]. AGU Annual Meeting 2024, Washington, D.C., USA. **https://agu.confex.com/agu/agu24/meetingapp.cgi/Paper/1598214**

2016 **S. C. Carter**., Carmichael, R.H., Estes, M.G., McBarnes, M. (2016, February 22). American Horseshoe Crab (Limulus polyphemus): Population ecology within the Northern Gulf of Mexico [Poster presentation, Session ME14D-0645]. AGU Ocean Sciences Meeting 2016, New Orleans, LA, USA. **https://agu.confex.com/agu/os16/meetingapp.cgi/Paper/88401**

CORE TECHNICAL SKILLS

Programming & Analysis: Python, SQL, R, GPU programming
Remote Sensing & GIS: Google Earth Engine, QGIS, Geopandas, Rasterio
Machine Learning: Deep Learning (UNET, CNNs), PyTorch, TensorFlow
Cloud Computing: AWS, Google Cloud Platform (GCS)
Specialized: Spatial Statistics, Agricultural Mapping, Auditory bird identification, classical music

AWARDS AND HONORS

Max Lanner Prize for Excellence in Instrumental Music. Colorado College (2015, 2016)

Languages

English: Native/Bilingual
Vietnamese: Limited working proficiency (C1)